

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: _____

July 4, 2014 Friday

Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} x^{\nu-1} e^{-x} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) In the inertial system S , an event is observed to take place at point A on the x -axis and 10^{-6} s later another event takes place at point B , 900m further down. Find the magnitude and direction of the velocity of S' with respect to S in which these two events appear simultaneous.

2. (20pts.) A pion moving along x -axis with $\beta = 0.8$ in the lab system decays by emitting a muon with $\beta' = 0.268$ along the y' -axis in the rest system of pion. Find the velocity ν of the muon in the lab system.

3. (20pts.) 3. Consider a monatomic ideal gas of N molecules.
- (a) What are the internal energy U and the equation of state of the ideal gas?
 - (b) Find the entropy S of this gas as the function of T and V , i.e., $S = S(T, V)$, setting (T_0, V_0) as the reference point where the entropy is S_0 .
 - (c) Find the Helmholtz free energy F of this gas as the function of T and V , i.e., $F = F(T, V)$.

4. (20pts.) Consider a gas in an adiabatically enveloped vessel of total volume V_f . The gas is initially at temperature T_i and confined by a divider to a volume V_i . Suppose that a divider is lifted, allowing the gas to abruptly and freely expand into final total volume V_f . At the end of this process, the gas reaches a temperature T_f .

(a) What is the change of the internal energy U in this process?

(b) Show that the temperature change in this process can be expressed as

$$\Delta T = T_f - T_i = - \int_{V_i}^{V_f} \frac{T^2}{C_V} \left(\frac{\partial(p/T)}{\partial T} \right)_V dV.$$

(c) If the gas obeys the Van der Waals equation, calculate the temperature change in this process (the C_V of Van der Waals gas is independent of V). Explain why the temperature change is positive or negative in terms of interaction among the molecules of the gas.

5. (20 pts.) Consider a rubber band of length l which is being stretched by external force f . Assume that change in volume of the rubber band due to the stretching is negligible.

(a) Write down the fundamental thermodynamic equation for the rubber band, analogous to the fundamental thermodynamic equation $dU = TdS - pdV$ for a system of fixed amount of gas.

(b) Show that $\left(\frac{\partial S}{\partial l} \right)_U = -\frac{f}{T}$.

(c) In an experiment, the length of the band is fixed at $l = 1$ m and the temperature of the rubber band is initially at $T = 300$ K and then is raised by a small increment $\Delta T = 3$ K. This causes the force needed to maintain the length of the band to increase by the amount of $\Delta f = 1.2$ N. In another experiment, the rubber band is stretched from $l = 1$ m by a small amount $\Delta l = 0.02$ m at constant temperature $T = 300$ K. What is the amount of the heat the band exchanges with the environment in the latter experiment? Is the heat released or absorbed by the band?

6. (20pts.) 6. The Maxwell velocity distribution describes the probability of a molecule of the ideal gas composed molecules of mass m and in equilibrium of temperature T in the velocity space. It states that the probability of finding a molecule with v_x between v_x and $v_x + dv_x$, v_y between v_y and $v_y + dv_y$, v_z between v_z and $v_z + dv_z$ is given by

$$P(v_x, v_y, v_z) dv_x dv_y dv_z = C \exp \left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right] dv_x dv_y dv_z$$

where C is a normalizing factor.

- Determine the normalizing factor C .
- Obtain the energy distribution $P(E) dE$, i. e. the probability of finding a molecule of with energy E between E and $E + dE$.
- What is the average energy $\langle E \rangle$ of the molecules?
- What is the uncertainty of the energy $\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2}$ of the molecules?